

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Chemistry

2. Subject Name: Inorganic quantitative analysis L-T-P: 0-0-6 Credits: 4

3. Pre-requisites:

4. Syllabus and reference books:

Syllabus:

Quantitative analysis: Redox titration (permanganate, dichromate), iodometric, complexometric and gravimetric titration (MnO_2 in pyrolusite, Cu^{2+} , Zn^{2+} , Ca^{2+} , Mg^{2+} in antacid, Ni^{2+} , Al^{3+} , Fe^{2+} and Fe^{3+} and their mixtures).

Determination of amount of calcium in a sample of milk powder by EDTA complexometry

Estimation of iodine in iodized salt by iodometric titration

Analysis of kidney stones by permanganometric titration

Reference Books:

(1) Vogel's Textbook of Quantitative Chemical Analysis, 4th edition, 1978

A. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Quantitative analysis: Redox titration (permanganate, dichromate), iodometric, complexometric and gravimetric titration (MnO_2 in pyrolusite, Cu^{2+} , Zn^{2+} , Ca^{2+} , Mg^{2+} in antacid, Ni^{2+} , Al^{3+} , Fe^{2+} and Fe^{3+} and their mixtures). Determination of amount of calcium in a sample of milk powder by EDTA complexometry Estimation of iodine in iodized salt by iodometric titration Analysis of kidney stones by permanganometric titration	
Total number of hours		80 h